Mean Squared Error (MSE)

θ_0	θ_1
450	0.15

		$h_\theta(x^i)$	y^i	$(h_\theta(x^i) - y^i)$	$(h_\theta(x^i) - y^i)^2$
1	2104		400		
1	1416		232		
1	1534		315		
1	852		178		

$$\frac{1}{2 \times 4}\sum_{i=1}^{4}(h_\theta(x^i) -$$

Root Mean Squared Error (RMSE)

θ_0	θ_1
450	0.15

		$h_\theta(x^i)$	y^i	$(h_\theta(x^i) - y^i)$	$(h_\theta(x^i) - y^i)^2$
1	2104		400		
1	1416		232		
1	1534		315		
1	852		178		

$$\frac{1}{4}\sum_{i=1}^{4}(h_\theta(x^i) - y^i)^2$$

$$J(\theta_0, \theta_1) = \sqrt{\frac{1}{2 \times 4}\sum_{i=1}^{4}(h_\theta(x^i) - y^i)^2}$$

Mean Absolute Error (MAE)

θ_0	θ_1
450	0.15

		$h_\theta(x^i)$	y^i	$(h_\theta(x^i) - y^i)$	$\lvert h_\theta(x^i) - y^i \rvert$
1	2104		400		
1	1416		232		
1	1534		315		
1	852		178		

$$\sum_{i=1}^{4} \lvert h_\theta(x^i) - y^i \rvert$$

$$\frac{1}{4}\sum_{i=1}^{4} \lvert h_\theta(x^i) - y^i \rvert$$

Gradient Descent – Single Iteration

θ_0	θ_1
450	0.15

		$h_\theta(x^i)$	y^i	$(h_\theta(x^i) - y^i)$	$\frac{\partial}{\partial\theta_0}J(\theta_0,\theta_1)$ $\frac{1}{4}\sum_{i=1}^{4}(h_\theta(x^i) - y^i)$	$\frac{\partial}{\partial\theta_1}J(\theta_0,\theta_1)$ $\frac{1}{4}\sum_{i=1}^{4}(h_\theta(x^i) - y^i).x^i$	$\alpha \times \frac{\partial}{\partial\theta_0}J(\theta_0,\theta_1)$	$\theta_0 - \alpha \times \frac{\partial}{\partial\theta_0}J(\theta_0,\theta_1)$
1	2104		400				$\alpha \times \frac{\partial}{\partial\theta_1}J(\theta_0,\theta_1)$	$\theta_1 - \alpha \times \frac{\partial}{\partial\theta_1}J(\theta_0,\theta_1)$
1	1416		232					
1	1534		315					
1	852		178					

